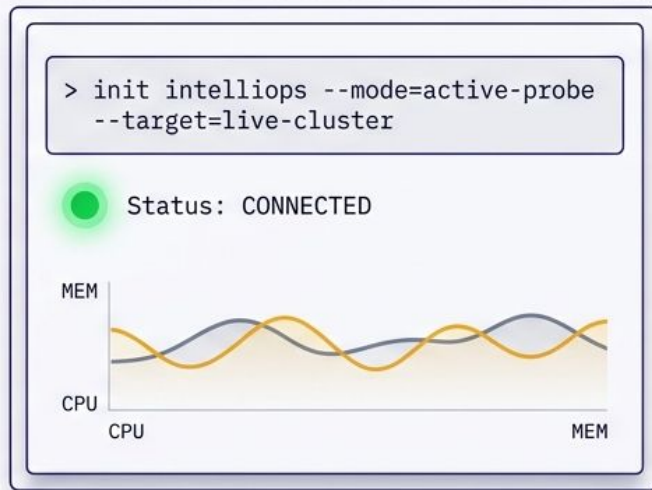


Active Cloud Diagnostics with IntelliOPS

Deploying a Virtual Engineer to eliminate manual RCA and accelerate incident resolution.



The Limit of Passive Observability



Step 1: Cloud incident triggers.



Step 2: SRE/Platform teams alerted (often off-hours).



Step 3: Engineers manually search historical logs and knowledge bases.

METRIC 01

Increased MTTR (Mean Time to Resolution).

METRIC 02

High Operational Overhead.

METRIC 03

Elevated Risk of Extended Downtime.

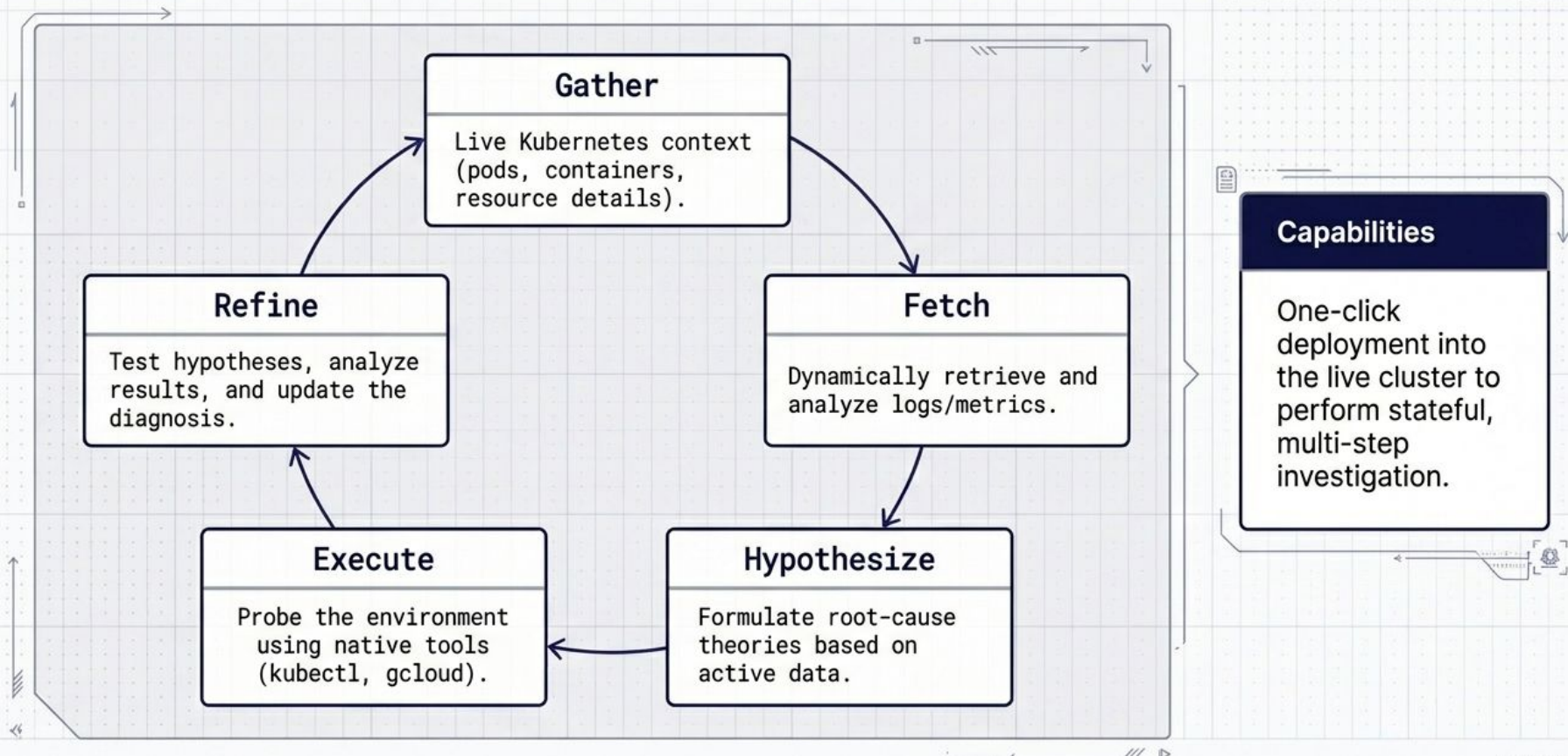
Existing tools observe historical data. They do not actively investigate live systems.

A Shift in Operational Stance

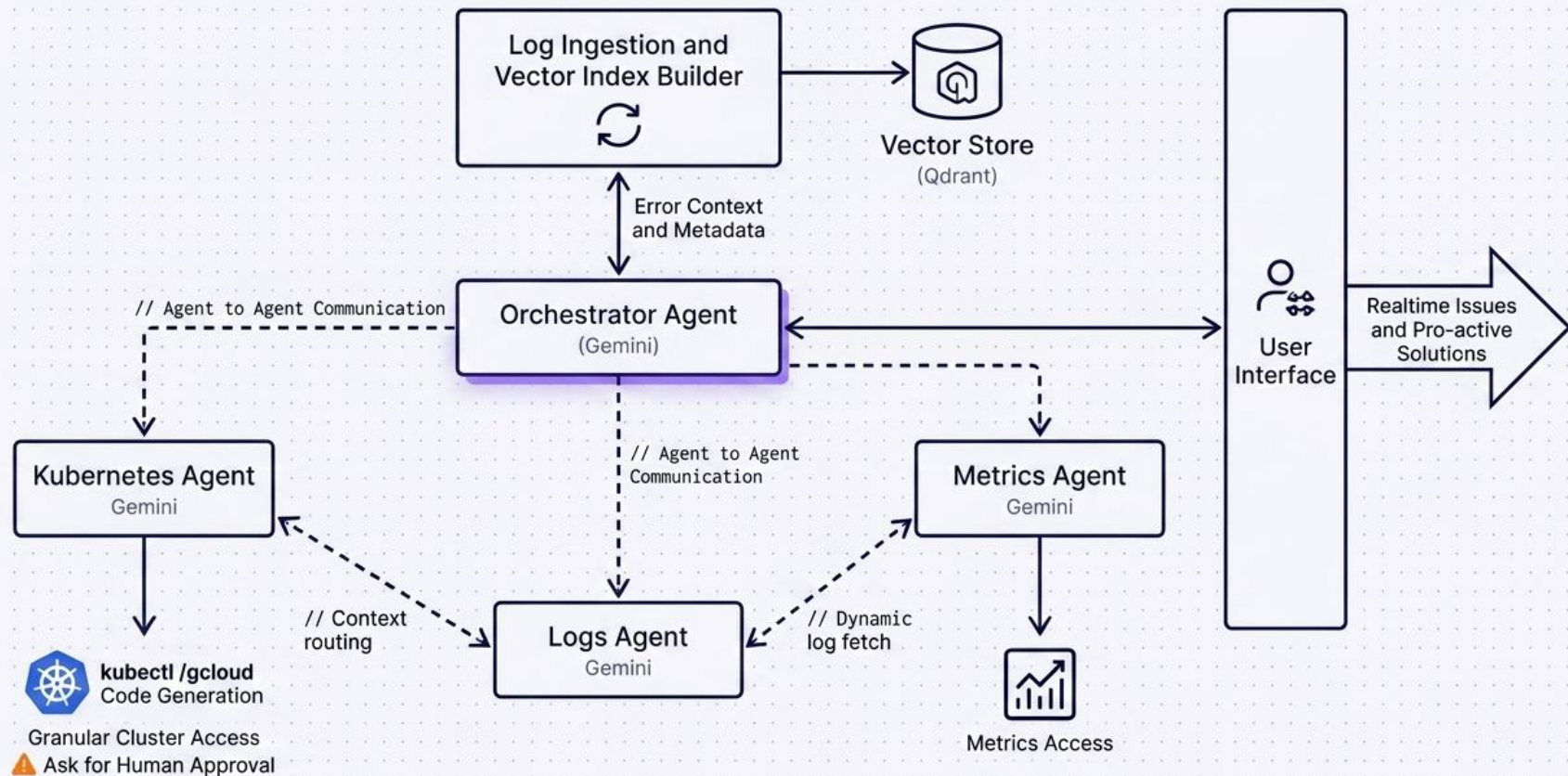
	Passive Observability	Active Virtual Engineering
Investigation Stance	Historical data search	Live cluster probing
Execution	Manual log hunting	Automated hypothesis testing via kubectl & gcloud
Resolution Speed	Human-dependent, bottlenecked	AI-accelerated, parallel processing
Remediation	Static playbooks	Context-aware action recommendation with Human-in-the-Loop

IntelliOPS acts as a Virtual Engineer, mirroring human troubleshooting logic in real-time.

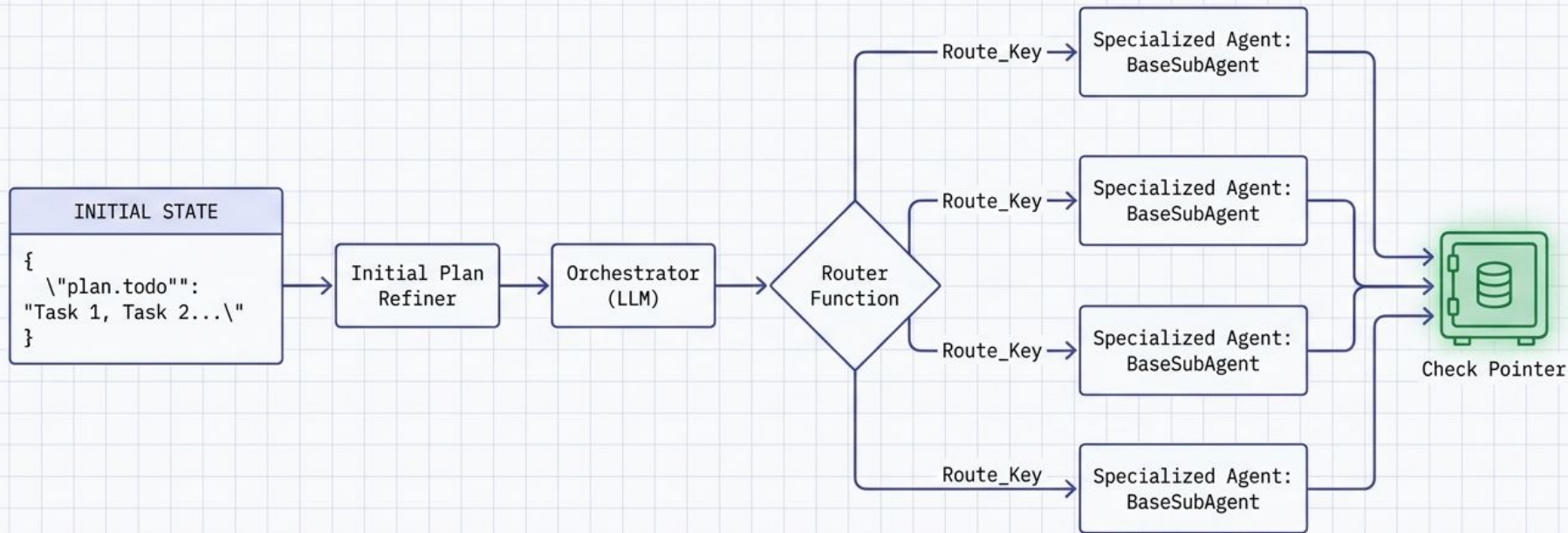
The Interactive Reasoning Loop



The Multi-Agent Constellation

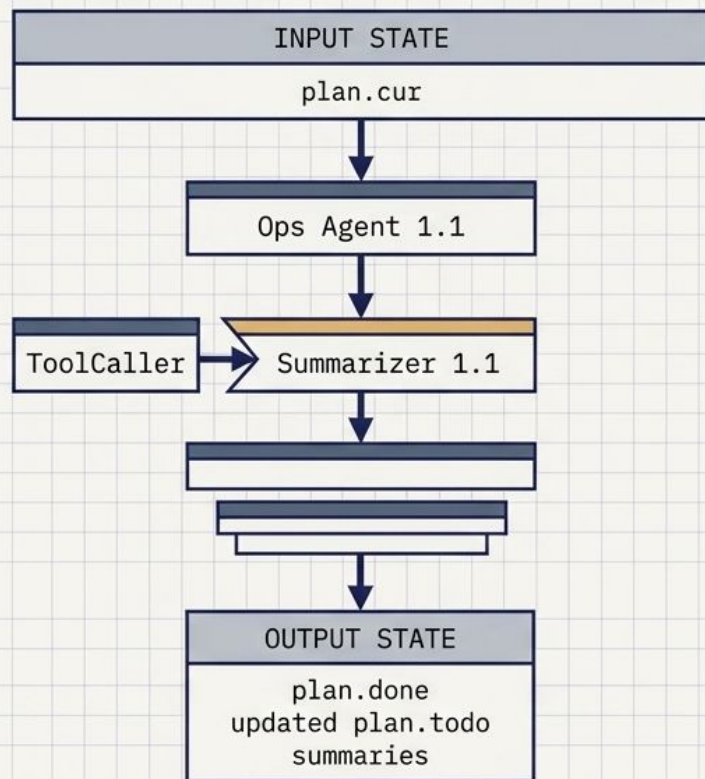


Stateful Orchestration via LangGraph



Complex incidents are broken down and routed dynamically. Stateful checkpoints ensure the agent never loses context mid-investigation.

Token Management & Plan Distillation



To prevent prompt growth and hallucination during iterative feedback, the system distills plans.

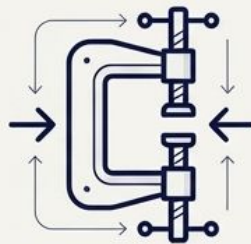
Task objects are dynamically managed, and a 'Conciser Node' compresses tool outputs to retain only the essential reasoning state.

Enterprise-Grade System Guardrails



High-Volume Noise Control

Uses standalone **shell script generation** and **statistical summarization** of metrics instead of passing raw, noisy time-series data.



Context Window Management

Employs the **Plan Distillation** architecture to break tasks into bite-sized objects and dynamically cycle the active context.



Hallucination Prevention

Integrates iterative error-feedback query refinement and a **Conciser Node** to strip irrelevant data and lock in accurate reasoning states.

Intelligent Grouping & Safe Remediation



Memory & Prioritization

- **LSH-Based Grouping:** Matches live anomalies with similar past incidents.
- **Causality-Based Dependency Resolution:** Prioritizes the sequence of issue handling to target root causes first.



[HITL]

The Human Gate

Human-in-the-Loop Execution.

IntelliOPS generates a clear RCA report and recommends corrective actions.

It **NEVER** executes remediation without explicit engineer approval.

The Anatomy of Resolution

Technical Domain

Live cluster
probing &
Multi-agent
parallelism

LSH Grouping &
Causality
Resolution

Context
Distillation &
HITL Gate

Vector-based
similarity
search (Qdrant)

Faster MTTR

(Drastic reduction in
diagnostic time)

Lower Support Cost

(Minimizes L3/on-call
burden and service
disruption)

Safer Remediation

(Actionable, verifiable
intelligence)

Improved Reliability

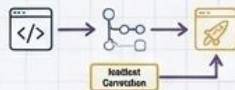
(The operational
knowledge base
strengthens over time)

Business Domain

Future Roadmap & Ecosystem Integration

Pipeline Correlation

Integrating with **CI/CD** and **GitHub Actions** to **correlate** live incidents with recent codebase changes for approval-based remediation.



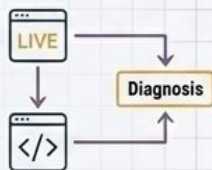
ITSM Alignment

Deep integration with **JIRA** and **ServiceNow** to seamlessly manage incident workflows.



Live Codebase Connection

Connecting directly to the live codebase to generate stronger, code-level diagnosis and fix recommendations.



Environment Expansion

Extending the IntelliOPS architecture to fully support On-prem environments.



The IntelliOPS Engineering Team

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[// SYSTEM: Active | STATUS: Ready]